



University of Asia Pacific

Department of Computer Science & Engineering

Course Title : Microprocessors and Microcontrollers Lab

Course Code : CSE 316

Report No : 06

Report Title : Remote AC Bulb Control

Date : 25 - 05 - 2026

Submitted By :

Name : Md. Osman Goni

Reg. No : 23101012

Section : A1

Submitted To :

Khadiza Sultana Sayma

Lecturer, Department of CSE

University Of Asia Pacific.

1. Experiment Name

Remote AC Bulb Control

2. Objective

To control an AC bulb wirelessly using an IR remote, Arduino, a relay module, and an IR receiver, so the bulb can be switched ON or OFF at a distance.

3. Apparatus / Hardware & Software Requirements

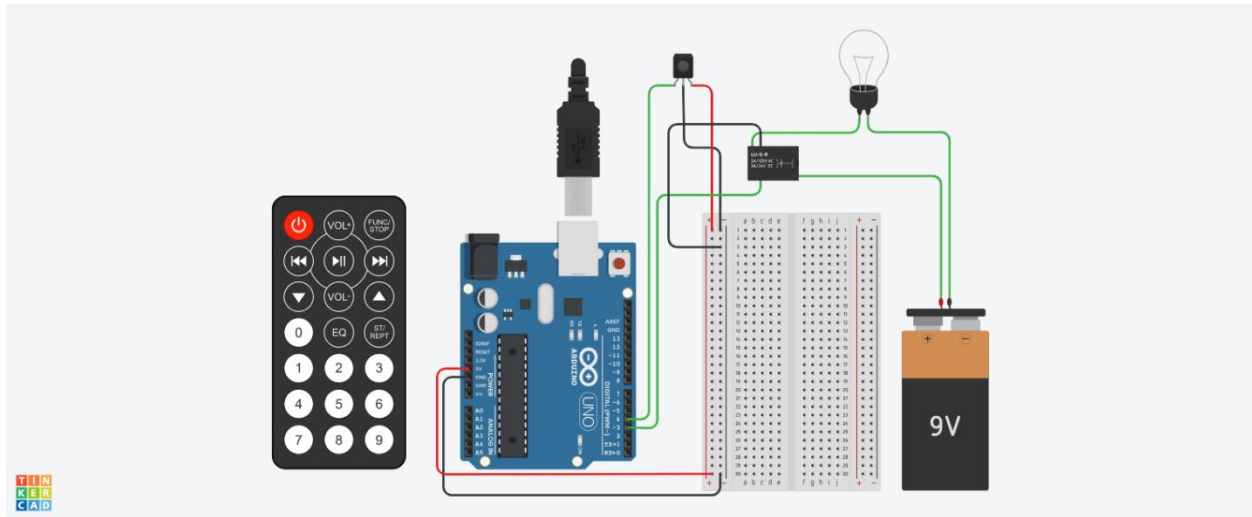
Hardware:

- Arduino UNO board
- IR Receiver sensor
- IR Remote (as shown, standard NEC protocol remote)
- Relay Module (suitable for AC voltage)
- AC Bulb with holder and wiring
- 9V Battery for relay/bulb
- Breadboard and connecting wires
- USB Cable for Arduino

Software:

- Arduino IDE
- IRremote library for Arduino (`#include <IRremote.hpp>`)

4. Circuit Diagram / Schematic



4. Code / Assembly Program

```
#include <IRremote.hpp>
const int relayPin = 3;
const int rcvPin = 4;
void setup() {
  pinMode(relayPin, OUTPUT);
  digitalWrite(relayPin, LOW);
  Serial.begin(9600);
  IrReceiver.begin(rcvPin,
    ENABLE_LED_FEEDBACK);
  Serial.println("IR receiver
  ready...");
}
void loop() {
  if (IrReceiver.decode()) {
```

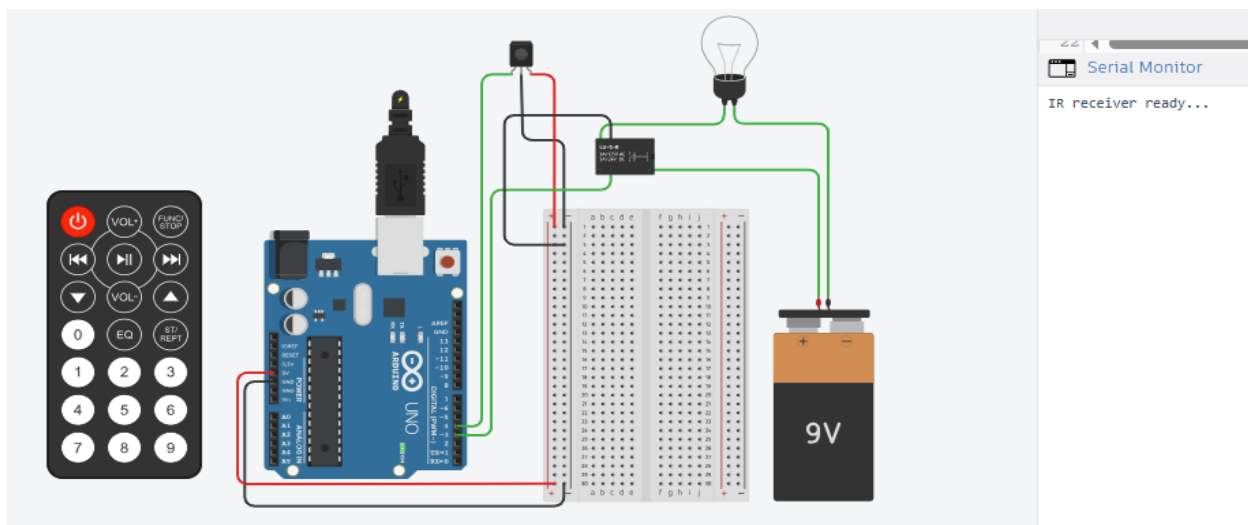
```

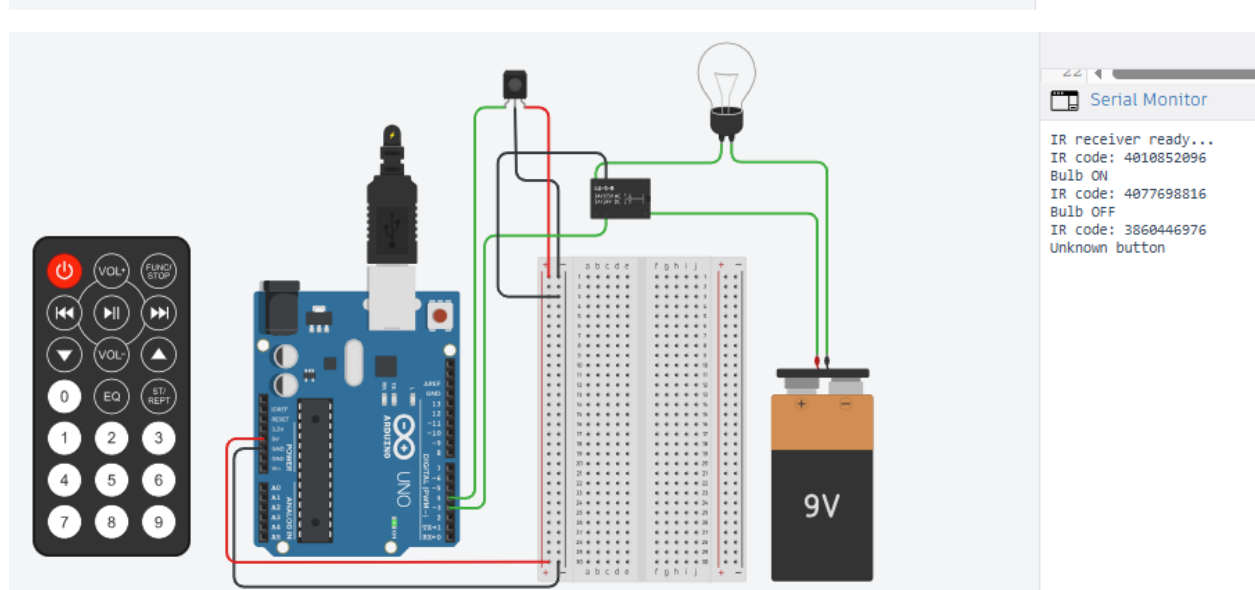
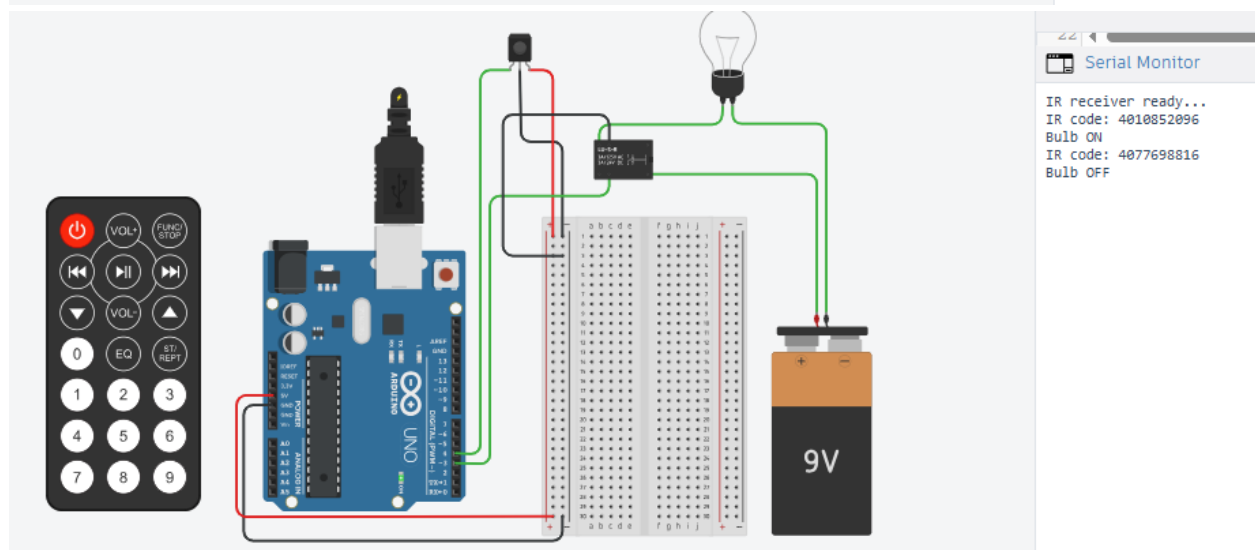
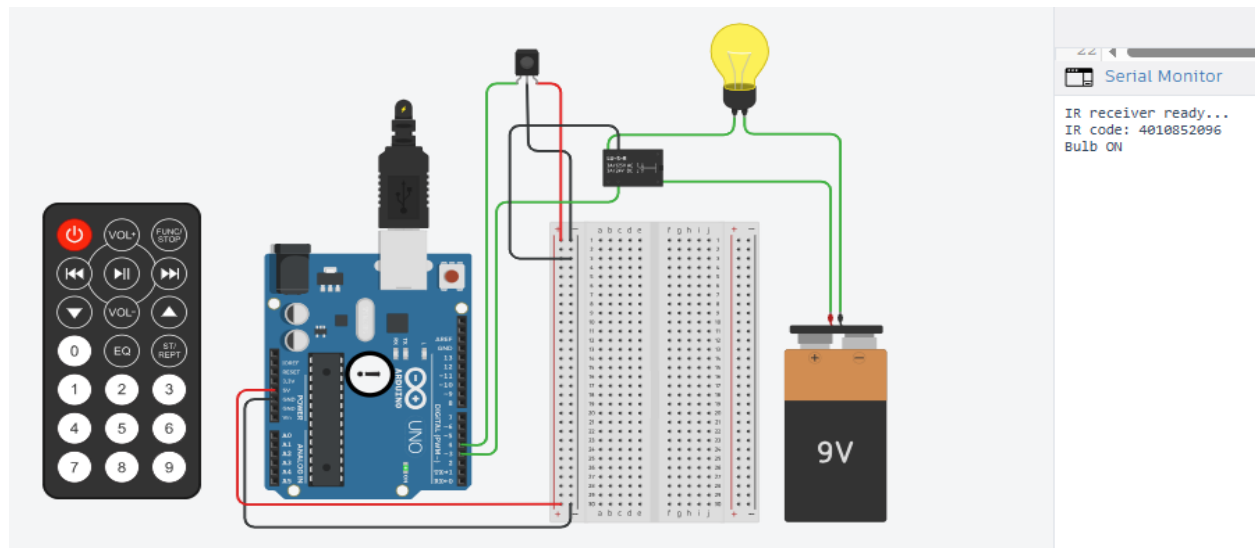
    unsigned long value =
IrReceiver.decodedIRData.decodedRawData;
    Serial.print("IR code: ");
    Serial.println(value);
    switch (value) {
        // ON button (replace with
your remote code)
        case 4010852096:
            Serial.println("Bulb ON");
            digitalWrite(relayPin,
HIGH);
            break;
        // OFF button
        case 4077698816:
            Serial.println("Bulb OFF");
            digitalWrite(relayPin, LOW);
            break;
        default:
            Serial.println("Unknown
button");
            break;
    }
    IrReceiver.resume();
}
delay(200);
}

```

5. Output / Observations

- On startup, Serial Monitor displays:
IR receiver ready...
- When a remote button is pressed, the Arduino prints the received value:
IR code: <number>
- If the received code matches a stored ON/OFF code, it prints:
 - Bulb ON and sets relay pin **HIGH**
 - Bulb OFF and sets relay pin **LOW**
- If a different button is pressed, it prints:
Unknown button





7. Result

After implementing and testing the system, the following observations were recorded:

- Pressing the programmed **ON button** (code 4010852096) energized the relay and the **AC bulb turned ON**.
- Pressing the programmed **OFF button** (code 4077698816) de-energized the relay and the **AC bulb turned OFF**.
- Other buttons produced “Unknown button” until their codes were added to the switch cases.
- The system successfully demonstrated reliable remote switching, confirmed by Serial Monitor outputs and the bulb’s response.

8. Conclusion

This project successfully implemented **remote AC load control** using an IR remote and Arduino. It demonstrated IR signal decoding with the IRremote library and safe switching of higher-voltage loads using a relay. The Serial Monitor provided a clear method to capture remote button codes and map them to ON/OFF actions.

With minor enhancements (adding more button functions, debouncing/repeat-code handling, and using a proper relay module enclosure), the design can be expanded into a practical home automation-style controller.